REPORTING

Report Date:	JUN-09-2023
Receipt Date:	JUN-05-2023
Collection Date:	JUN-02-2023
Specimen:	Blood
Status:	FINAL

PHYSICIAN

Chih-Hsueh Chen
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Address: F15., No 207-5 Sec 3, Beixin Rd, Xindian Dist, New Taipei City, 23143, Taiwan
Ph: +886 963 820 633 | Fax: N/A
Additional Recipient: N/A

Complete Tumor Response Map on page 2

Summary of Detected Somatic Alterations, Immunotherapy Biomarkers & Associated Treatment Options

KEY ✔ Approved in indication ~ Approved in other indication ✕ Lack of response

Detected Alteration(s) / Biomarker(s)	Associated FDA-approved therapies	Clinical trial availability (see page 3)	% cfDNA or Amplification
EGFR E746_A750del (Exon 19 deletion)	✔ **Afatinib, Dacomitinib, Erlotinib, Erlotinib+ramucirumab, Gefitinib, Osimertinib**	Yes	93.9%
PIK3CA H1047R	~ Alpelisib	Yes	0.5%
TP53 G266V	None	Yes	80.1%
EGFR Amplification	None	Yes	High (+++)
CCNE1 Amplification	None	Yes	Medium (++)
PIK3CA Amplification	None	Yes	High (+++)

Additional Biomarkers

Biomarker	Additional Details
MSI-High	NOT DETECTED

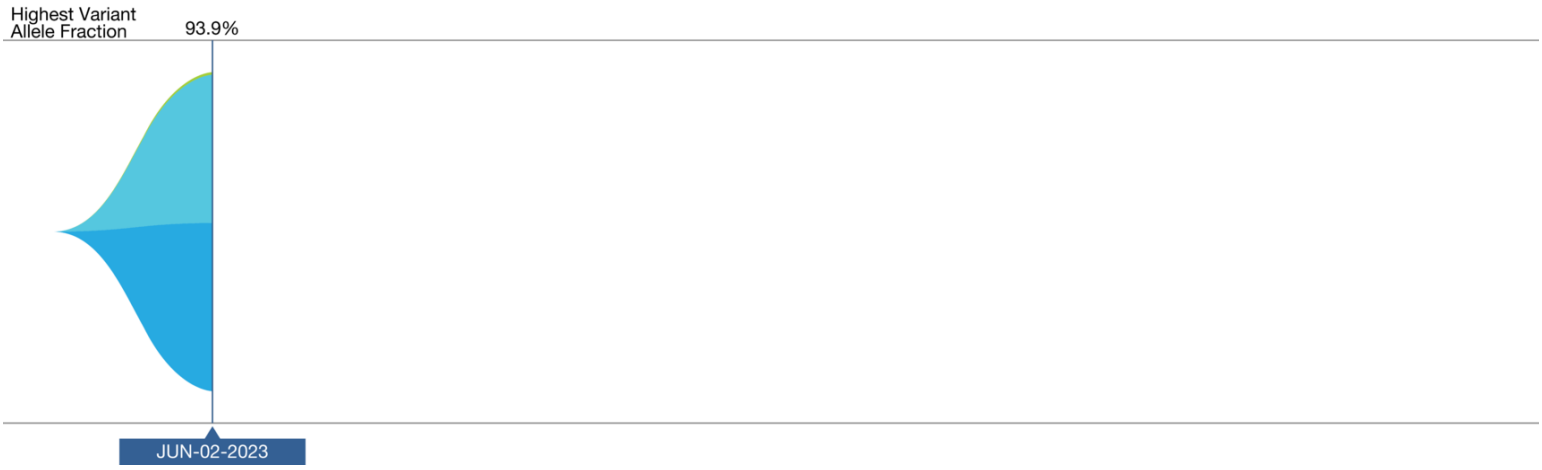
Alterations or biomarkers that were "NOT DETECTED" have been excluded from the summary table above.

We evaluated this sample for 74 genes, including the following guideline-recommended genes for NSCLC

EGFR(T790M and others) | *ALK* | *ROS1* | *BRAF* | *MET* | *ERBB2*(HER2) | *RET* | *NTRK* | *KRAS*

Guardant360 Tumor Response Map

The Guardant360 Tumor Response Map illustrates the variant allele fraction (% cfDNA) of observed somatic variants at each sample submission. Amplifications are not plotted, and only the first and last five test dates are plotted. Please see the Physician Portal (portal.guardanthealth.com) for the Tumor Response Map with all test dates.



Detected Alteration(s) / Biomarker(s)	% cfDNA or Amp
EGFR E746_A750del (Exon 19 deletion)	93.9%
TP53 G266V	80.1%
PIK3CA H1047R	0.5%
EGFR Amplification Amplifications not graphed above	High (+++) Plasma Copy Number 18.9
CCNE1 Amplification Amplifications not graphed above	Medium (++) Plasma Copy Number 5.1
PIK3CA Amplification Amplifications not graphed above	High (+++) Plasma Copy Number 4.1

The table above annotates the variant allele fraction (% cfDNA) detected in this sample, listed in descending order.
§ See definitions section for more detail

Available Clinical Trials (within the same state as the ordering physician)

There may be additional trials not listed here. Visit: portal.guardanthealth.com or email clientservices@guardanthealth.com with A0771511 in the subject line of the email, for additional trials.

Alteration	Trial ID / Contact	Title	Phase	Site(s)
EGFR E746_A750del	NCT04077463 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib as Monotherapy or in Combination With Amivantamab in Participants With Advanced Non-small Cell Lung Cancer	Phase 1	Kaohsiung, Taiwan Taipei City, Taiwan Tainan, Taiwan Taichung, Taiwan
	NCT04988295 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Amivantamab and Lazertinib in Combination With Platinum-Based Chemotherapy Compared With Platinum-Based Chemotherapy in Patients With Epidermal Growth Factor Receptor (EGFR)-Mutated Locally Advanced or Metastatic Non- Small Cell Lung Cancer After Osimertinib Failure	Phase 3	New Taipei City, Taiwan Kaohsiung, Taiwan Taipei, Taiwan Tainan, Taiwan Additional trial sites available
	NCT05215548 Jin-Shing Chen, M.D., Ph.D.,chenjs@ntu.edu.tw,886-2-2322-0322	Primary Tumor Resection With EGFR TKI for Stage IV NSCLC	Phase 2	Taipei, Taiwan (2)
	NCT05388669 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib With Subcutaneous Amivantamab Compared With Intravenous Amivantamab in Participants With Epidermal Growth Factor Receptor (EGFR)-Mutated Advanced or Metastatic Non-small Cell Lung Cancer	Phase 3	Kaohsiung City, Taiwan Kaohsiung, Taiwan Taipei City, Taiwan Taipei, Taiwan Additional trial sites available
	NCT05442060 Anna Hu,annahu@obipharma.com,886-2-27866589 x104	To Evaluate OBI-833/OBI-821 in Combination With First-Line Erlotinib in Patients With EGFR-Mutated, Globo H-Positive, Locally Advanced or Metastatic Non-Small Cell Lung Cancer	Phase 2	Taoyuan, Taiwan Taichung, Taiwan Taipei, Taiwan (4)
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
PIK3CA H1047R	NCT03337698 Reference Study ID Number: BO39610 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	A Study Of Multiple Immunotherapy-Based Treatment Combinations In Participants With Metastatic Non-Small Cell Lung Cancer (Morpheus- Non-Small Cell Lung Cancer)	Phase 1 /Phase 2	Taipei City, Taiwan
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
EGFR Amplification	NCT04077463 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib as Monotherapy or in Combination With Amivantamab in Participants With Advanced Non-small Cell Lung Cancer	Phase 1	Kaohsiung, Taiwan Taipei City, Taiwan Tainan, Taiwan Taichung, Taiwan
	NCT04988295 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Amivantamab and Lazertinib in Combination With Platinum-Based Chemotherapy Compared With Platinum-Based Chemotherapy in Patients With Epidermal Growth Factor Receptor (EGFR)-Mutated Locally Advanced or Metastatic Non- Small Cell Lung Cancer After Osimertinib Failure	Phase 3	New Taipei City, Taiwan Kaohsiung, Taiwan Taipei, Taiwan Tainan, Taiwan Additional trial sites available
	NCT05099172 Bayer Clinical Trials Contact,clinical-trials-contact@bayer.com,(+)1-888-84 22937	First in Human Study of BAY2927088 in Participants Who Have Advanced Non-small Cell Lung Cancer (NSCLC) With Mutations in the Genes of Epidermal Growth Factor Receptor (EGFR) and/or Human Epidermal Growth Factor Receptor 2 (HER2)	Phase 1	Taipei, Taiwan Tainan, Taiwan Taoyuan, Taiwan Taichung, Taiwan
	NCT05215548 Jin-Shing Chen, M.D., Ph.D.,chenjs@ntu.edu.tw,886-2-2322-0322	Primary Tumor Resection With EGFR TKI for Stage IV NSCLC	Phase 2	Taipei, Taiwan (2)
	NCT05388669 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib With Subcutaneous Amivantamab Compared With Intravenous Amivantamab in Participants With Epidermal Growth Factor Receptor (EGFR)-Mutated	Phase 3	Kaohsiung City, Taiwan Kaohsiung, Taiwan Taipei City, Taiwan Taipei, Taiwan

Alteration	Trial ID / Contact	Title	Phase	Site(s)
		Advanced or Metastatic Non-small Cell Lung Cancer		Additional trial sites available
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
PIK3CA Amplification	NCT03337698 Reference Study ID Number: BO39610 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	A Study Of Multiple Immunotherapy-Based Treatment Combinations In Participants With Metastatic Non-Small Cell Lung Cancer (Morpheus- Non-Small Cell Lung Cancer)	Phase 1 /Phase 2	Taipei City, Taiwan
	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
TP53 G266V	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			
CCNE1 Amplification	Visit portal.guardanthealth.com for trials not within the same state as the physician's office			

More clinical trial options available at portal.guardanthealth.com

Definitions

Amplification: Guardant360 detects amplifications in the genes listed in Table 1. Gene amplification results in increased copies of the gene present in the cfDNA. The reported absolute copy number value represents the average copy number for the detected gene that was detected in circulating cfDNA. With the exception of sex-linked genes such as *AR*, 2 copies are expected in the absence of amplification. As the absolute number of copies in circulation is dependent on both tumor fraction and the magnitude of the tumor amplification, amplifications are reported on a semi-quantitative scale.
For *CCNE1*, *EGFR* and *FGFR1*, three levels are reported:
Low (+): Amplification magnitude is below the 50th percentile of amplifications detected by Guardant360.
Medium (++): Amplification magnitude is between the 50th and 90th percentiles.
High (+++): Amplification magnitude is above the 90th percentile.

For *BRAF*, *CCND1*, *CCND2*, *CDK4*, *CDK6*, *ERBB2*, *FGFR2*, *KIT*, *KRAS*, *MET*, *PDGFRA*, *RAF1*, *MYC*, *PIK3CA* and *AR*, two levels are reported:
Medium (++): Amplification magnitude is below the 50th percentile of amplifications detected by Guardant360.
High (+++): Amplification magnitude is above the 50th percentile.

Deletion (Del): The following alteration was detected in this patient: *EGFR* E746_A750del. Guardant360 detects short deletions in exons of certain genes (see Table 1), including potential splice site-disrupting events.

Interpretation

Somatic alterations were detected in the circulating cell-free DNA isolated from this patient's blood specimen. These genomic alterations are cancer-associated somatic variants, some of which have been associated with either increased or reduced clinical response to specific treatments. The percentage of altered cell-free DNA circulating (% cfDNA) in blood is related to the unique tumor biology of each patient. Factors that may affect the % cfDNA of detected somatic alterations include tumor growth, turn over, size, heterogeneity, vascularization, disease progression, and treatment.

Amplification was detected in the circulating cell-free DNA isolated from this patient’s blood specimen for the annotated gene(s). Unlike tissue-based gene amplification tests (e.g. IHC or FISH), Guardant360 assesses the total representation of a given gene in all circulating cell-free DNA present in the patient’s blood sample including material derived from the tumor and healthy tissue alike. As such, the absolute level of amplification present in the blood depends both on the tumor-derived cfDNA content and on the degree of amplification within that fraction and cannot be inferred from bulk cfDNA interrogation. For example, a positive Guardant360 test could represent a small population of cells with extremely high levels of the detected gene amplification. Alternatively, it could represent a large population of cells with low to medium levels of the detected gene amplifications. The exact correlation between amplification detected by Guardant360 compared to IHC or FISH and how each test differentially guides patient management is an area of active investigation.

Method and Limitations

Guardant360 sequences 74 cancer-associated genes to identify somatic alterations. Cell-free DNA (cfDNA) is extracted from plasma, enriched for targeted regions, and sequenced using the Illumina platform and hg19 as the reference genome. All exons are sequenced in some genes; only clinically significant exons are sequenced in other genes. The types of genomic alterations detected by Guardant360 include single nucleotide variants, gene amplifications, fusions, short insertions/deletions (longest detected, 70 base pairs), and splice site disrupting events (see Table 1). Microsatellite Instability (MSI) is assessed for all cancer types by evaluating somatic changes in the length of repetitive sequences on the Guardant360 panel. A "Not Detected" result in samples where the highest % cfDNA is < 0.2% is an inconclusive result because it does not preclude MSI-High status in tissue. MSI status is currently not reported for earlier panel versions. This version of the Guardant360 test is not validated for the detection of other types of genomic alterations, such as complex rearrangements or gene deletions. Certain sample or variant characteristics, such as low cfDNA concentration, may result in reduced analytic sensitivity. Guardant360 cannot discern the source of circulating cfDNA, and for some variants in the range of ~40 to 60% cfDNA, the test cannot easily distinguish germline variants from somatic alterations. Guardant360 is not validated for the detection of germline or de novo variants that are associated with hereditary cancer risk. Tissue genotyping should be considered when plasma genotyping is negative, if clinically appropriate.

Table 1: Genes on the Guardant360 Panel

Guardant360 reports single nucleotide variants, splice site mutations, and insertion and deletion variants (indels) in all clinically relevant exons in 74 genes and reports other variant types in select genes as indicated below.

AKT1	*ALK* #	*APC*	*AR* †	*ARAF*	*ARID1A*	*ATM*	*BRAF* †	*BRCA1*
BRCA2	*CCND1* †	*CCND2* †	*CCNE1* †	*CDH1*	*CDK12*	*CDK4* †	*CDK6* †	*CDKN2A*
CTNNB1	*DDR2*	*EGFR* †	*ERBB2* †	*ESR1*	*EZH2*	*FBXW7*	*FGFR1* †	*FGFR2* † #
FGFR3 #	*GATA3*	*GNA11*	*GNAQ*	*GNAS*	*HNF1A*	*HRAS*	*IDH1*	*IDH2*
JAK2	*JAK3*	*KIT* †	*KRAS* †	*MAP2K1*	*MAP2K2*	*MAPK1*	*MAPK3*	*MET* †
MLH1	*MPL*	*MTOR*	*MYC* †	*NF1*	*NFE2L2*	*NOTCH1*	*NPM1*	*NRAS*
NTRK1 #	*NTRK3*	*PDGFRA* †	*PIK3CA* †	*PTEN*	*PTPN11*	*RAF1* †	*RB1*	*RET* #
RHEB	*RHOA*	*RIT1*	*ROS1* #	*SMAD4*	*SMO*	*STK11*	*TERT* ‡	*TP53*
TSC1	*VHL*							

‡ Guardant360 reports alterations in the promoter region of this gene.

Guardant360 reports fusion events involving this gene.

† Guardant360 reports amplifications of this gene.

About the Test

The Guardant360 assay was developed and its performance characteristics were determined by Guardant Health, Inc. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. This test may be used for clinical purposes and should not be regarded as investigational or for research only. Guardant Health's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing. The laboratory report should be interpreted in the context of other clinical information and laboratory, pathology, and imaging studies by a qualified medical professional prior to initiating or changing a patient's treatment plan. The selection of any, all, or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) is entirely at the discretion of the treating medical professional. Drug and trial information are based on the diagnosis written on the submitted test request form; this information is not based on any supplemental information provided by the requesting medical professional, including pathology reports or other molecular studies. Some drugs listed in this report may not be approved or cleared by the FDA for the indicated use. Guardant Health makes no endorsement, express or implied, of any product, physician, or procedure contained in this report. This report makes no promises or guarantees that a particular medication will affect (or not affect) the clinical outcome of any patient.

Testing Performed at: Guardant Health

Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive, Redwood City, CA, 94063, USA

Additional information is available

Any therapeutic annotations are based on publicly available information. This information is described in the "Detailed Therapy Results" and "Relevance of Detected Alterations" sections.

Visit **portal.guardanthealth.com** or email **clientservices@guardanthealth.com** with A0771511 in the subject line of the email for:

- Additional clinical trials
- Detailed Therapy Results
- Relevance of Detected Alterations
- References

If you would like to receive this additional information with every Guardant360 report, please call client services at **855.698.8887** to opt-in.

Additional information begins on the next page.

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
EGFR E746_A750del	NCT04077463 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib as Monotherapy or in Combination With Amivantamab in Participants With Advanced Non-small Cell Lung Cancer	Phase 1	Seattle, WA; Detroit, MI; Saint Louis, MO; Philadelphia, PA; Portland, OR; Salt Lake City, UT; Tampa, FL; Fairfax, VA; Boston, MA (3); New York, NY (2); CA (5); Puerto Rico; Japan (7); China (14); Taiwan (4); Korea, Republic of (4); Italy (5); France (7); Germany (8); Spain (8)
	NCT04951635 See https://clinicaltrials.gov/show/NCT04951635	A Phase III Study to Assess the Effects of Almonertinib Following Chemoradiation in Patients With Stage III Unresectable Non-small Cell Lung Cancer	Phase 3	China
	NCT04982900 Hecheng Li, PhD, MD, lihecheng2000@hotmail.com, +8613917113402	Treatment of EGFR-TKI for Residual Lesions of Multiple Synchronous Ground-glass Opacities	Phase 2	China (6)
	NCT04988295 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Amivantamab and Lazertinib in Combination With Platinum-Based Chemotherapy Compared With Platinum-Based Chemotherapy in Patients With Epidermal Growth Factor Receptor (EGFR)-Mutated Locally Advanced or Metastatic Non-Small Cell Lung Cancer After Osimertinib Failure	Phase 3	East Brunswick, NJ; Grand Island, NE; Wytheville, VA; Orange, CA; Colorado Springs, CO; Fort Lauderdale, FL; Memphis, TN; Los Angeles, CA; Tucson, AZ; Jackson, MS; Duarte, CA; Mobile, AL; Pittsburgh, PA; Puyallup, WA; Burlington, VT; Athens, GA; Cincinnati, OH; Horsham, PA; Fairfax, VA; Charlottesville, VA; Portland, OR (2); TX (5); Puerto Rico; Hong Kong; Denmark; Czechia (3); Malaysia (5); Portugal (4); Sweden (5); Netherlands (3); China (30); Poland (7); Korea, Republic of (9); Brazil (21); France (13); Bulgaria (4); Argentina (7); Japan (11); United Kingdom (6); Spain (17); India (8); Canada (3); Turkey (10); Belgium (7); Taiwan (7); Mexico (5); Israel (7); Germany (9)
	NCT05215548 Jin-Shing Chen, M.D., Ph.D.,chenjs@ntu.edu.tw,886-2-2322-0322	Primary Tumor Resection With EGFR TKI for Stage IV NSCLC	Phase 2	Taiwan (2)
	NCT05388669 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib With Subcutaneous Amivantamab Compared With Intravenous Amivantamab in Participants With Epidermal Growth Factor Receptor (EGFR)-Mutated Advanced or Metastatic Non-small Cell Lung Cancer	Phase 3	Denver, CO; Orlando, FL; Philadelphia, PA; East Brunswick, NJ; New York, NY; New Brunswick, NJ; Bronx, NY; Boca Raton, FL; Kansas City, KS; Duarte, CA; Long Beach, CA; Irvine, CA; Charleston, SC; Fairfax, VA; Ann Arbor, MI; Portland, OR (2); Argentina (6); Japan (18); United Kingdom (4); Malaysia (6); Thailand (3); Portugal (4); Spain (16); Canada (2); Turkey (11); China (32); Taiwan (8); Poland (5); Korea, Republic of (9); Brazil (16); Italy (9); Israel (5); France (4); Australia (6); Germany (6)
	NCT05442060 Anna Hu,annahu@obipharma.com,886-2-27866589 x104	To Evaluate OBI-833/OBI-821 in Combination With First-Line Erlotinib in Patients With EGFR-Mutated, Globo H-Positive, Locally Advanced or Metastatic Non-Small Cell Lung Cancer	Phase 2	Taiwan (6)

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
	NCT05507606 See https://clinicaltrials.gov/show/NCT05507606	Study of Osimertinib+Bevacizumab+Chemotherapy for EGFR+ Advanced Non-Small Cell Lung Cancer With Concurrent Mutations	Phase 2	China
	NCT05686434 chen chen,chen_checn@tmu.edu.cn, 13920761627	Osimertinib Therapy After Resection in High-risk Stage I EGFRm NSCLC (OSTAR)	Phase 2	China
	NCT05810350 Wei Wang, doctor,19574605@qq.com, 13635324483	Aumolertinib Combined Intrathecal Chemotherapy for Leptomeningeal Metastasis From EGFR-Mutated NSCLC and Prognostic Value of Dynamic Changes in cfDNA Profiles	Phase 2	China
PIK3CA H1047R	NCT03006172 Reference Study ID Number: GO39374 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	To Evaluate the Safety, Tolerability, and Pharmacokinetics of Inavolisib Single Agent in Participants With Solid Tumors and in Combination With Endocrine and Targeted Therapies in Participants With Breast Cancer	Phase 1	Nashville, TN; Boston, MA (2); New York, NY (2); Canada; France; United Kingdom (3); Spain (2)
	NCT03337698 Reference Study ID Number: BO39610 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	A Study Of Multiple Immunotherapy-Based Treatment Combinations In Participants With Metastatic Non-Small Cell Lung Cancer (Morpheus- Non-Small Cell Lung Cancer)	Phase 1 /Phase 2	Cleveland, OH; Newark, DE; Taiwan; Australia; Korea, Republic of (2); United Kingdom (3); Israel (3); France (5); Spain (7)
	NCT04586335 Jason Sudia, PhD, MPH,jason.sudia@haihepharma.com,9083801329	Study of CYH33 in Combination With Olaparib an Oral PARP Inhibitor in Patients With Advanced Solid Tumors.	Phase 1	China
	NCT05216432 Relay Therapeutics Inc, ClinicalTrials@relaytx.com,617-322-0731	First-in-Human Study of Mutant-selective PI3Kα Inhibitor, RLY-2608, as a Single Agent in Advanced Solid Tumor Patients and in Combination With Fulvestrant in Patients With Advanced Breast Cancer	Phase 1	Denver, CO; Tucson, AZ; Orlando, FL; Houston, TX; Madison, WI; Boston, MA; Plantation, FL; Nashville, TN; Fairfax, VA; New York, NY (2); Spain (2)
	NCT05300048 Study Inquiry, clinicaltrials@faeththerapeutics.com,(708) 406-9282	Combination of Serabelisib and Insulin Suppressing Diet With or Without Nab-paclitaxel in Subjects With Advanced Solid Tumors With PIK3CA Mutations	Phase 1	Philadelphia, PA; Sioux Falls, SD; New York, NY; Greenville, NC; Los Angeles, CA; Houston, TX; Newport Beach, CA; Rochester, MN; Kingwood, TX; Indianapolis, IN; Beaumont, TX; Anaheim, CA; Dallas, TX; Birmingham, AL; Lake Success, NY
	NCT05307705 Patient Advocacy, clinicaltrials@loxooncology.com;,855-569-6305	A Study of LOXO-783 in Patients With Breast Cancer/Other Solid Tumors	Phase 1	Houston, TX; Saint Louis, MO; Rochester, MN; Atlanta, GA; New York, NY; San Antonio, TX; Nashville, TN; Santa Monica, CA; Boston, MA (2); Dallas, TX (2); Singapore; Canada (2); Japan (4); Australia (2)
EGFR Amplification	NCT03574402 Yi-Long Wu, Professor,syylwu@live.cn, 862083827812	Phase II Umbrella Study Directed by Next Generation Sequencing	Phase 2	China
	NCT04077463 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib as Monotherapy or in Combination With Amivantamab in Participants With Advanced Non-small Cell Lung Cancer	Phase 1	Seattle, WA; Detroit, MI; Saint Louis, MO; Philadelphia, PA; Portland, OR; Salt Lake City, UT; Tampa, FL; Fairfax, VA; Boston, MA (3); New York, NY (2); CA (5); Puerto Rico; Japan (7); China (14); Taiwan (4); Korea, Republic of (4); Italy (5); France (7); Germany (8); Spain (8)
	NCT04982900 Hecheng Li, PhD, MD, lihecheng2000@hotmail.com,	Treatment of EGFR-TKI for Residual Lesions of Multiple Synchronous Ground-glass Opacities	Phase 2	China (6)

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
	+8613917113402			
	NCT04988295 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Amivantamab and Lazertinib in Combination With Platinum-Based Chemotherapy Compared With Platinum-Based Chemotherapy in Patients With Epidermal Growth Factor Receptor (EGFR)-Mutated Locally Advanced or Metastatic Non-Small Cell Lung Cancer After Osimertinib Failure	Phase 3	East Brunswick, NJ; Grand Island, NE; Wytheville, VA; Orange, CA; Colorado Springs, CO; Fort Lauderdale, FL; Memphis, TN; Los Angeles, CA; Tucson, AZ; Jackson, MS; Duarte, CA; Mobile, AL; Pittsburgh, PA; Puyallup, WA; Burlington, VT; Athens, GA; Cincinnati, OH; Horsham, PA; Fairfax, VA; Charlottesville, VA; Portland, OR (2); TX (5); Puerto Rico; Hong Kong; Denmark; Czechia (3); Malaysia (5); Portugal (4); Sweden (5); Netherlands (3); China (30); Poland (7); Korea, Republic of (9); Brazil (21); France (13); Bulgaria (4); Argentina (7); Japan (11); United Kingdom (6); Spain (17); India (8); Canada (3); Turkey (10); Belgium (7); Taiwan (7); Mexico (5); Israel (7); Germany (9)
	NCT05099172 Bayer Clinical Trials Contact,clinical-trials-contact@bayer.com,(+)1-888-84 22937	First in Human Study of BAY2927088 in Participants Who Have Advanced Non-small Cell Lung Cancer (NSCLC) With Mutations in the Genes of Epidermal Growth Factor Receptor (EGFR) and/or Human Epidermal Growth Factor Receptor 2 (HER2)	Phase 1	Houston, TX; Detroit, MI; Bethesda, MD; Boston, MA; Atlanta, GA; Nashville, TN; Fairfax, VA; Gilbert, AZ; Duarte, CA (2); Singapore (2); Hong Kong (2); Japan (9); Portugal (2); Spain (6); Netherlands (2); Belgium (2); China (12); Taiwan (4); Poland (2); Korea, Republic of (6); Brazil (3); Italy (8); Israel (2); France (5)
	NCT05215548 Jin-Shing Chen, M.D., Ph.D.,chenjs@ntu.edu.tw,886-2-2322-0322	Primary Tumor Resection With EGFR TKI for Stage IV NSCLC	Phase 2	Taiwan (2)
	NCT05388669 Study Contact,Participate-In-This-Study@its.jnj.com,844-434-4210	A Study of Lazertinib With Subcutaneous Amivantamab Compared With Intravenous Amivantamab in Participants With Epidermal Growth Factor Receptor (EGFR)-Mutated Advanced or Metastatic Non-small Cell Lung Cancer	Phase 3	Denver, CO; Orlando, FL; Philadelphia, PA; East Brunswick, NJ; New York, NY; New Brunswick, NJ; Bronx, NY; Boca Raton, FL; Kansas City, KS; Duarte, CA; Long Beach, CA; Irvine, CA; Charleston, SC; Fairfax, VA; Ann Arbor, MI; Portland, OR (2); Argentina (6); Japan (18); United Kingdom (4); Malaysia (6); Thailand (3); Portugal (4); Spain (16); Canada (2); Turkey (11); China (32); Taiwan (8); Poland (5); Korea, Republic of (9); Brazil (16); Italy (9); Israel (5); France (4); Australia (6); Germany (6)
PIK3CA Amplification	NCT03006172 Reference Study ID Number: GO39374 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	To Evaluate the Safety, Tolerability, and Pharmacokinetics of Inavolisib Single Agent in Participants With Solid Tumors and in Combination With Endocrine and Targeted Therapies in Participants With Breast Cancer	Phase 1	Nashville, TN; Boston, MA (2); New York, NY (2); Canada; France; United Kingdom (3); Spain (2)
	NCT03337698 Reference Study ID Number: BO39610 https://forpatients.roche.com/,global-roche-genentech-trials@gene.com,888-662-6728 (U.S. and Canada)	A Study Of Multiple Immunotherapy-Based Treatment Combinations In Participants With Metastatic Non-Small Cell Lung Cancer (Morpheus- Non-Small Cell Lung Cancer)	Phase 1 /Phase 2	Cleveland, OH; Newark, DE; Taiwan; Australia; Korea, Republic of (2); United Kingdom (3); Israel (3); France (5); Spain (7)
	NCT04192981 Brandon Imber, MD, PhD,imberb@mskcc.org,6312126346	GDC-0084 With Radiation Therapy for People With PIK3CA-Mutated Solid Tumor Brain Metastases or Leptomeningeal Metastases	Phase 1	Basking Ridge, NJ; Uniondale, NY; Harrison, NY; Seattle, WA; Miami, FL; Montvale, NJ; Commack, NY;

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
				New York, NY; Middletown, NJ (2)
	NCT04586335 Jason Sudia, PhD, MPH,jason.sudia@haihepharma.com,9083801329	Study of CYH33 in Combination With Olaparib an Oral PARP Inhibitor in Patients With Advanced Solid Tumors.	Phase 1	China
	NCT05216432 Relay Therapeutics Inc, ClinicalTrials@relaytx.com,617-322-0731	First-in-Human Study of Mutant-selective PI3Kα Inhibitor, RLY-2608, as a Single Agent in Advanced Solid Tumor Patients and in Combination With Fulvestrant in Patients With Advanced Breast Cancer	Phase 1	Denver, CO; Tucson, AZ; Orlando, FL; Houston, TX; Madison, WI; Boston, MA; Plantation, FL; Nashville, TN; Fairfax, VA; New York, NY (2); Spain (2)
	NCT05300048 Study Inquiry, clinicaltrials@faeththerapeutics.com,(708) 406-9282	Combination of Serabelisib and Insulin Suppressing Diet With or Without Nab-paclitaxel in Subjects With Advanced Solid Tumors With PIK3CA Mutations	Phase 1	Philadelphia, PA; Sioux Falls, SD; New York, NY; Greenville, NC; Los Angeles, CA; Houston, TX; Newport Beach, CA; Rochester, MN; Kingwood, TX; Indianapolis, IN; Beaumont, TX; Anaheim, CA; Dallas, TX; Birmingham, AL; Lake Success, NY
TP53 G266V	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04085315 Lisa Tan,Lisa.Tan@ucsf.edu,(415) 353-7710	Alisertib in Combination With Osimertinib in Metastatic EGFR-mutant Lung Cancer	Phase 1	San Francisco, CA
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
	NCT04869475 Min Shi, MD & Ph. D,sm11998@rjh.com.cn,+86-21-64370045	Arsenic Trioxide in Refractory Solid Tumors With Rescuable p53 Mutation	Phase 2	China
	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
CCNE1 Amplification	NCT04553133 Pfizer CT.gov Call Center,ClinicalTrials.gov_Inquiries@pfizer.com,1-800-718-1021	PF-07104091 as a Single Agent and in Combination Therapy	Phase 1 /Phase 2	Houston, TX; Grand Rapids, MI; Clive, IA; Middletown, NJ; Des Moines, IA (3); Boston, MA (3); Charlottesville, VA (3); NY (9); KY (11); Japan (2); China (2); Mexico (3)
	NCT05238922 Incyte Corporation Call Center (US), medinfo@incyte.com,1.855.463.3463	Study of INCB123667 in Subjects With Advanced Solid Tumors	Phase 1	Duarte, CA; Philadelphia, PA; Huntersville, NC; Atlanta, GA; Pittsburgh, PA; New Haven, CT; Irvine, CA; Lone Tree, CO; Fort Worth, TX; New York, NY (2); Netherlands (2); Japan (5); United Kingdom (4); Italy (5); France (5); Switzerland (3)
	NCT05252416 Blueprint Medicines, medinfo@blueprintmedicines.com,617-714-6707	(VELA) Study of BLU-222 in Advanced Solid Tumors	Phase 1 /Phase 2	Houston, TX; New York, NY; Chicago, IL; Sarasota, FL; Little Rock, AR; Charlottesville, VA; Boston, MA (2)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
PIK3CA Amplification H1047R	ABTL0812		Inhibitor of mTORC1/mTORC2 /Dhfr.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Pancreatic carcinoma, Endometrial carcinoma)
	Alpelisib	Piqray	p-110-alpha inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (HR+, HER2-) with a PIK3CA mutation, as determined by a validated test)
	Apitolisib		Dual PI3K/mTOR inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Prostate carcinoma, Endometrial carcinoma, Renal cell carcinoma, Breast carcinoma)
	Archexin	Archexin	Akt1 antisense oligonucleotide.	Phase 1 (Solid Tumor) Phase 2 (Pancreatic carcinoma, Renal cell carcinoma)
	ARQ 751		Akt inhibitor.	Phase 1 (Solid Tumor)
	ASN003		Dual Braf and PI3K inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	AT13148		Multi-AGC (Akt, p70S6K) small molecule kinase inhibitor.	Phase 1 (Solid Tumor)
	Bimiralisib		Dual PI3K/mTOR inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Breast carcinoma (triple negative), Primary central nervous system lymphoma (PCNSL))
	Buparlisib		PI3K inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Head and neck squamous cell carcinoma (HNSCC), Breast carcinoma)
	Capivasertib		Akt inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Prostate carcinoma, Breast carcinoma)
	CC-115		DNA-PK/dual mTORC1/2 kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Glioblastoma, Glioma)
	Copanlisib	Aliqopa	PI3K inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Follicular lymphoma (FL))
	CYH33		p110-alpha-specific small molecule inhibitor.	Phase 1 (Solid Tumor)
	Fimepinostat		PI3K/HDAC inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Thyroid carcinoma, Diffuse large B-cell lymphoma (DLBCL))
	Gedatolisib		Dual PI3K/mTOR inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Breast carcinoma)
	HS-10352		p110-alpha-specific small molecule inhibitor.	Phase 1 (Breast carcinoma)
	Inavolisib		PI3K inhibitor.	Phase 1 (Solid Tumor)
	Ipatasertib		Akt inhibitor.	Phase 1 (Solid Tumor) Phase 3 (Prostate carcinoma, Breast carcinoma)
	ME-344		Dual mTORC1/mTORC2 inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Ovarian carcinoma, Small cell lung carcinoma (SCLC))
	Miransertib		Akt inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	MK-2206		Akt inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Gastric carcinoma, Gallbladder carcinoma, Hepatocellular carcinoma (HCC), Pancreatic carcinoma, Hodgkin lymphoma (HL), Lymphoma, Melanoma, Carcinoid tumor, Prostate carcinoma, Endometrial carcinoma, Renal cell carcinoma, Ovarian carcinoma, Small cell lung carcinoma (SCLC), Adenoid cystic carcinoma, Head and neck carcinoma, Gastroesophageal junction carcinoma, Gastrointestinal neuroendocrine carcinoma, Mucinous colon adenocarcinoma, Breast carcinoma, Chronic lymphocytic leukemia (CLL), Acute myeloid leukemia (AML), Cholangiocarcinoma, Thymic neoplasm, Peritoneal papillary serous carcinoma, Lung cancer, Colorectal carcinoma (CRC), Diffuse large B-cell lymphoma (DLBCL))
	MSC2363318A		Akt1, Akt3, and p70S6K inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Lymphoma)
	Onatasertib		Dual mTORC1/mTORC2 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Solid Tumor, Bladder neuroendocrine carcinoma, Multiple myeloma (MM), Diffuse large B-cell lymphoma (DLBCL))
	Paxalisib		Dual PI3K/mTOR inhibitor.	Phase 2 (Glioblastoma, Breast carcinoma)
	Perifosine		Akt inhibitor, induces apoptosis; mechanism of action is context specific.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Multiple myeloma (MM), Colorectal carcinoma (CRC))
	Pilaralisib		PI3K inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Endometrial carcinoma, Breast carcinoma)
	RLY-2608		p110-alpha-specific small molecule inhibitor.	Phase 1 (Solid Tumor)
	RMC-5552		mTORC1-specific inhibitor.	Phase 1 (Solid Tumor)
	Samotolisib		Dual PI3K/mTOR inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Pancreatic carcinoma, Prostate carcinoma)
	Sapanisertib		Dual mTORC1/mTORC2 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Uterine carcinosarcoma, Hepatocellular carcinoma (HCC), Lymphoma, Merkel cell carcinoma, Anaplastic thyroid carcinoma, Pancreatic neuroendocrine tumor, Prostate carcinoma, Endometrial carcinoma, Renal cell carcinoma, Ovarian carcinoma, Urothelial carcinoma, Bladder carcinoma, Breast carcinoma, Multiple myeloma (MM), Non-Hodgkin lymphoma (NHL), Lung cancer, Sarcoma, Acute lymphoblastic leukemia (ALL))
	Serabelisib		PI3K inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Endometrial carcinoma, Renal cell carcinoma, Breast carcinoma (triple negative))
	SF1126		Dual PI3K/mTOR inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Head and neck squamous cell carcinoma (HNSCC))
	TAS-117		Akt inhibitor.	Phase 2 (Solid Tumor)
	TAS0612		Akt/p70S6K/p90RSK1 multikinase inhibitor.	Phase 1 (Solid Tumor)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	Triciribine		DNA synthesis and Akt inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Ovarian carcinoma, Breast carcinoma)
	Uprosertib		Akt inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Melanoma, Uveal melanoma, Breast carcinoma, Cervical carcinoma, Acute myeloid leukemia (AML), Multiple myeloma (MM))
	Vistusertib		Dual mTORC1/mTORC2 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Glioblastoma, Meningioma, Small cell lung carcinoma (SCLC), Solid Tumor, Diffuse large B-cell lymphoma (DLBCL))
EGFR E746_A750del (Exon 19 deletion)	Aumolertinib		Egfr T790M inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC))
	Avitinib		Irreversible mutation-specific Egfr kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Non-Hodgkin lymphoma (NHL))
	AZD3759		Egfr tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	BBT-176		Fourth generation Egfr inhibitor targeting exon 19del/L858R, T790M, and C797S mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	BDTX-1535		Irreversible brain-penetrant fourth generation Egfr inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Glioma)
	BLU-451		Egfr inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	BLU-701		Fourth generation Egfr inhibitor targeting exon 19del, L858R, and C797X resistance mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	BLU-945		Fourth generation Egfr inhibitor targeting T790M and T790M /C797S mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	BPI-361175		Fourth generation Egfr inhibitor targeting T790M and T790M /C797S mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	BPI-7711		Egfr T790M inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	CLN-081		Covalent mutation-specific (L858R, T790M, exon 19 deletion, exon 20 insertion) Egfr tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Dacomitinib	Vizimpro	Pan-ErbB family tyrosine kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (EGFR-mutant lung carcinoid, EGFR-mutant lung large cell neuroendocrine carcinoma, EGFR-mutant NSCLC)
	DBPR112		Egfr tyrosine kinase inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Head and neck squamous cell carcinoma (HNSCC))
	Erlotinib	Tarceva	Egfr tyrosine kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
				indications (EGFR-mutant lung carcinoid, EGFR-mutant lung large cell neuroendocrine carcinoma, NSCLC with EGFR exon 19 del/L858R, Pancreatic carcinoma, EGFR-mutant NSCLC)
	Erlotinib+bevacizumab	Tarceva+Avastin		Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Lung cancer)
	Erlotinib+ramucirumab	Tarceva+Cyramza	Egfr tyrosine kinase inhibitor + anti-VEGFR-2 monoclonal antibody combination.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (NSCLC with EGFR exon 19 del /L858R)
	Gefitinib	Iressa	Egfr tyrosine kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (EGFR-mutant lung carcinoid, EGFR-mutant lung large cell neuroendocrine carcinoma, EGFR-mutant NSCLC)
	H002		Fourth generation Egfr inhibitor targeting exon 19del/L858R, T790M, and C797S mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Hemay022		Egfr tyrosine kinase inhibitor.	Phase 1 (Breast carcinoma (HER2+))
	Icotinib	Conmana	Egfr inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Esophageal carcinoma)
	JIN-A02		Fourth generation Egfr inhibitor targeting T790M and T790M /C797S mutations.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Lazertinib		Third generation mutation-specific Egfr tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Mobocertinib	Exkivity	Mutation-specific Egfr/Her2 inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (NSCLC with EGFR exon 20 insertion)
	Naquotinib		EGFR mutant-specific inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	Nazartinib		Third generation EGFR mutant-specific (T790M, L858R, exon 19 deletion) tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	NX-019		Egfr inhibitor.	Phase 1 (Solid Tumor)
	Olafertinib		Third generation mutation-specific (T790M, L858R, exon 19 deletion) Egfr tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	Olmutinib		Egfr inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Osimertinib	Tagrisso	Egfr T790M inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (EGFR-mutant lung carcinoid, EGFR-mutant lung large cell neuroendocrine carcinoma, EGFR-mutant NSCLC)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	PF-06747775		Egfr T790M-specific inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	TAS3351		Fourth generation Egfr inhibitor targeting T790M and T790M /C797S mutations.	
	ZN-e4		Egfr T790M inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
EGFR Amplification	ABBV-221		Anti-Egfr antibody drug conjugate.	Phase 1 (Solid Tumor)
	ABBV-321		Anti-Egfr antibody conjugated to monomethyl auristatin F.	Phase 1 (Solid Tumor) Phase 1 (Glioblastoma, Head and neck squamous cell carcinoma (HNSCC), Brain and Central Nervous System Tumors, Lung squamous cell carcinoma)
	ABT806		Anti-Egfr and EGFRvIII antibody.	Phase 1 (Solid Tumor) Phase 2 (Gastroesophageal junction carcinoma)
	ASP-1929		An antibody-dye conjugate comprised of cetuximab and IRDye700DX acting as photoimmunotherapy.	Phase 3 (Head and neck carcinoma)
	AVID100		Anti-Egfr antibody-drug conjugate with DM1.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Head and neck squamous cell carcinoma (HNSCC), Breast carcinoma (triple negative))
	BCA101		Anti-Egfr antibody fused to the extracellular domain of human TGF-beta receptor II.	Phase 1 (Solid Tumor) Phase 1 (Head and neck squamous cell carcinoma (HNSCC), Brain and Central Nervous System Tumors, Anal squamous cell carcinoma)
	CMAB009		Anti-Egfr monoclonal antibody.	Phase 3 (Colorectal carcinoma (CRC))
	CPGJ602		Recombinant human-mouse chimeric anti-Egfr monoclonal antibody.	Phase 2 (Colorectal carcinoma (CRC))
	D2C7-IT		Immunotoxin targeting both wild-type Egfr and Egfr-vIII.	Phase 1 (Glioblastoma)
	EGFR(V)-EDV-Dox		Doxorubicin-loaded EGFR-targeting nanocells.	Phase 1 (Glioblastoma)
	GC1118		Anti-Egfr monoclonal antibody.	Phase 1 (Solid Tumor) Phase 2 (Gastric carcinoma, Glioblastoma, Gastroesophageal junction carcinoma)
	KL-140		Anti-Egfr monoclonal antibody.	Phase 3 (Colorectal carcinoma (CRC))
	MM-151		Anti-Egfr monoclonal antibody mixture.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	Nimotuzumab	Theraloc	Anti-Egfr monoclonal antibody.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Gastric carcinoma, Glioblastoma, Glioma, Pancreatic carcinoma, Head and neck carcinoma, Gastroesophageal junction carcinoma, Esophageal carcinoma, Cervical carcinoma)
	REGN7075		anti-EGFR/CD28 bispecific antibody.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Head and neck squamous cell

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
				carcinoma (HNSCC), Cutaneous squamous cell carcinoma, Breast carcinoma (triple negative), Colorectal carcinoma (CRC))
	SI-B001		Anti-Egfr/ErbB3 bispecific antibody.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Head and neck squamous cell carcinoma (HNSCC), Esophageal carcinoma)
	Sym004		Anti-Egfr antibody mixture.	Phase 2 (Glioma, Head and neck squamous cell carcinoma (HNSCC), Colorectal carcinoma (CRC))
	SYN004		Anti-Egfr monoclonal antibody.	Phase 1 (Solid Tumor) Phase 1 (Lung squamous cell carcinoma)
	TAK-186		A Conditional Bispecific Redirected Activation (COBRA) Protein targeting Egfr and CD3.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Head and neck squamous cell carcinoma (HNSCC), Colorectal carcinoma (CRC))
EGFR Amplification E746_A750del (Exon 19 deletion)	Afatinib	Gilotrif	Irreversible pan-ErbB kinase inhibitor.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Lung squamous cell carcinoma, EGFR-mutant NSCLC)
	Amivantamab	Rybrevant	Bispecific anti-Met/Egfr antibody.	Phase 3 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (NSCLC with EGFR exon 20 insertion)
	BAY2927088		Egfr/Her2 kinase inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC))
	BBP-398		Shp-2 inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor)
	BDTX-189		Irreversible Egfr/Her2 inhibitor.	Phase 2 (Solid Tumor)
	ERAS-601		Shp-2 inhibitor.	Phase 1 (Solid Tumor)
	Furmonertinib		Third generation mutation-specific (T790M, L858R, exon 19 deletion) Egfr tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC))
	Lifirafenib		Dual Braf/Egfr inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Solid Tumor, Brain and Central Nervous System Tumors)
	Neratinib	Nerlynx	Egfr/Her2/ErbB4 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Breast carcinoma (HER2+))
	Pirotinib		ErbB family inhibitor.	Phase 1 (Solid Tumor)
	Poziotinib		Egfr/Her2/ErbB4 kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Gastric carcinoma, Head and neck squamous cell carcinoma (HNSCC), Breast carcinoma, Esophageal squamous cell carcinoma, Colorectal carcinoma (CRC))
	Pyrotinib		Egfr/Her2 kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Breast carcinoma)
	SKLB1028		Egfr/Flt3/c-Abl inhibitor.	Phase 2 (Acute myeloid leukemia (AML))
	Sunvozertinib		Bispecific anti-Egfr/Her2 monoclonal antibody.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Non-Hodgkin lymphoma (NHL))
	TAS2940		Egfr/Her2 kinase inhibitor.	Phase 1 (Solid Tumor)
	Varlitinib		Egfr/Her2 kinase inhibitor.	Phase 2 (Gastric carcinoma, Hepatocellular carcinoma (HCC), Pancreatic carcinoma, Cholangiocarcinoma)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	WSD0922-FU		Blood-brain barrier penetrable EGFR/EGFRvIII inhibitor.	Phase 1 (Non-small cell lung carcinoma (NSCLC)) Phase 1 (Glioblastoma, Anaplastic astrocytoma)
TP53 G266V	Adavosertib		Wee1 tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Embryonal tumor with multi-layered rosettes (ETMR), Medulloblastoma, Small cell lung carcinoma (SCLC), Solid Tumor, Primary myelofibrosis (PMF), Ovarian carcinosarcoma, Acute myeloid leukemia (AML), MDS/MPN, unclassifiable, Chronic myelomonocytic leukemia (CMML), Peritoneal papillary serous carcinoma, Myelodysplastic Syndrome (MDS))
	Alisertib		Aurora A kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 3 (Peripheral T-cell lymphoma (PTCL))
	ALT-801		p53-targeted T-cell receptor-IL2 fusion.	Phase 1 (Solid Tumor) Phase 2 (Melanoma, Urothelial carcinoma, Bladder carcinoma, Urethral melanoma, Urethral carcinoma, Multiple myeloma (MM))
	AMG 900		Aurora A/B/C kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Acute myeloid leukemia (AML))
	AT9283		Aurora A, B, Jak2, Jak3, Bcr-Abl kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Acute myeloid leukemia (AML), Multiple myeloma (MM), Acute lymphoblastic leukemia (ALL))
	ATO	Trisenox	PML-RARA inhibitor. Inhibits multiple signaling pathways, including the Hedgehog pathway.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) FDA Approved in other indications (Acute myeloid leukemia (AML), Acute promyelocytic leukemia (APL))
	AZD2811		Nanoparticle formulation of Aurora kinase B inhibitor barasertib (AZD1152).	Phase 1 (Solid Tumor) Phase 2 (Acute myeloid leukemia (AML), Myelodysplastic Syndrome (MDS))
	COTI-2		Reactivates mutant p53.	Phase 1 (Endometrial carcinoma, Head and neck squamous cell carcinoma (HNSCC), Ovarian carcinoma, Cervical carcinoma)
	Debio 0123		Wee1 tyrosine kinase inhibitor.	Phase 1 (Solid Tumor)
	ENMD-2076		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Fibrolamellar hepatocellular carcinoma, Ovarian carcinoma, Breast carcinoma, Fallopian tube adenocarcinoma, Sarcoma)
	Eprenetapopt		Reactivates mutant p53.	Phase 3 (Myelodysplastic Syndrome (MDS))
	Kevetrin		Blocks Mdm2-p53 interaction, restoring transcriptional activity of p53.	Phase 1 (Solid Tumor) Phase 2 (Ovarian carcinoma)
	LY3295668		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Head and neck squamous cell carcinoma (HNSCC), Small cell lung carcinoma (SCLC), Breast carcinoma (triple negative), Breast carcinoma (hormone receptor +, HER2-))
	SGT-53		TP53 gene therapy delivered via transferrin-targeted nanoparticles.	Phase 1 (Solid Tumor) Phase 2 (Glioblastoma, Glioma, Pancreatic carcinoma)
	SNS-314		Aurora A/B kinase inhibitor.	Phase 1 (Solid Tumor)
	TAS-119		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	TT-00420		Aurora A/B kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Breast carcinoma (triple negative))
CCNE1 Amplification	Adavosertib		Wee1 tyrosine kinase inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Embryonal tumor with multi-layered rosettes (ETMR), Medulloblastoma, Small cell lung carcinoma (SCLC), Solid Tumor, Primary myelofibrosis (PMF), Ovarian carcinosarcoma, Acute myeloid leukemia (AML), MDS/MPN, unclassifiable, Chronic myelomonocytic leukemia (CMML), Peritoneal papillary serous carcinoma, Myelodysplastic Syndrome (MDS))
	BLU-222		Cdk2 inhibitor.	Phase 2 (Solid Tumor)
	Dinaciclib		Cdk inhibitor targeting several Cdks, including Cdk1, Cdk2, Cdk5, and Cdk9.	Phase 2 (Lung adenocarcinoma) Phase 3 (Chronic lymphocytic leukemia (CLL))
	INCB0123667		Cdk2 inhibitor.	Phase 1 (Solid Tumor)
	PF-07104091		Cdk2 inhibitor.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Ovarian carcinoma, Small cell lung carcinoma (SCLC), Breast carcinoma)
	Seliciclib		Cdk inhibitor targeting several Cdks, including Cdk2, Cdk5, Cdk7, and Cdk9.	Phase 2 (Non-small cell lung carcinoma (NSCLC)) Phase 2 (Nasopharyngeal carcinoma, Lung cancer)
PIK3CA H1047R	LOXO-783		p110-alpha H1047R inhibitor.	

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
EGFR E746_A750del	The presence of an EGFR abnormality (mutation, amplification, or overexpression) can result in an overabundance or overactivity of Egfr protein, which can lead to excessive proliferation. [1].	The presence of a sensitizing EGFR mutation in a tumor is the strongest biological predictor of sensitivity to an Egfr tyrosine kinase inhibitor (TKI). Compared with conventional chemotherapy, Egfr TKIs have been shown to improve progression-free survival in non-small cell lung cancer patients whose tumors harbor EGFR mutations. [2-5]. The Egfr TKIs erlotinib, afatinib, gefitinib, osimertinib, and dacomitinib have been approved by the FDA for the treatment of non-small cell lung cancer (NSCLC) with exon 19 deletion or L858R EGFR mutations; osimertinib has additionally been approved for the treatment of NSCLC with EGFR T790M. [2,5-11]. Afatinib has additionally been FDA-approved for the treatment of NSCLC with S768I, L861Q, and/or G719X mutations. [12]. The combination of erlotinib and ramucirumab has been FDA-approved for the treatment of metastatic NSCLC patients with tumors harboring an EGFR exon 19 deletion or the exon 21 L858R mutation. [13]. Amivantamab and mobocertinib have been approved by the FDA for NSCLC patients with EGFR exon 20 insertions, whose disease has progressed on or after platinum-based chemotherapy. [14-17]. Studies have reported non-squamous NSCLC patients with metastatic disease and tumors harboring an EGFR exon 19 deletion or L858R mutation to be sensitive to osimertinib, erlotinib, afatinib, gefitinib, dacomitinib, and the combination of erlotinib plus ramucirumab. [2,5-7,11,13,18]. Less common activating EGFR mutations have variable sensitivity to EGFR tyrosine kinase inhibitors. [19].	Some patients with EGFR-mutant NSCLC exhibit resistance to Egfr inhibition; resistance has been associated with insertions in EGFR exon 20, the T790M mutation in EGFR, and amplification of either MET or ERBB2. [20-24]. Third generation irreversible Egfr TKIs that target the EGFR T790M mutation have shown efficacy in T790M-mutant NSCLC, including osimertinib, which has received approval by the FDA, EMA, and PMDA for the treatment of EGFR T790M-mutant metastatic NSCLC. [25-29]. Several studies have reported that resistance to Egfr TKIs in NSCLC is mediated by the transformation of NSCLC cell types to those of SCLC with neuroendocrine features. [30-33]. Preclinical studies have reported increased Smo expression in NSCLC cell lines resistant to first, second, and third generation Egfr inhibitors as compared with sensitive ones; treatment with Smo inhibitors was observed to restore sensitivity in the resistant cell lines. [34-36].
PIK3CA H1047R	PIK3CA mutations are not mutually exclusive with EGFR or KRAS or BRAF mutations, and are associated with increased PI3K signaling and increased activation of Akt. [37,38]. PIK3CA mutations have been associated with activation of PI3K/Akt signaling and colony formation in NSCLC cell lines, and the PIK3CA H1047R activating mutation has been shown to drive tumorigenesis in combination with BRAF V600E in a mouse model of NSCLC. [37,39].	Activating PIK3CA alterations may predict sensitivity to PI3K/Akt/mTOR pathway inhibitors, several of which have been approved in specific cancer indications, including the PI3K inhibitors alpelisib and copanlisib, and the mTOR inhibitors everolimus and temsirolimus. [40-42]. While PIK3CA activating alterations have been suggested to predict sensitivity to the mTOR inhibitors everolimus and temsirolimus, results from clinical studies have been mixed, with several reporting no associations between PIK3CA mutational status and response to therapy. [41,43-46]. Other	PIK3CA mutations have been reported in EGFR-mutant NSCLC patients following emergence of resistance to osimertinib and have been associated with osimertinib resistance in preclinical NSCLC models. Combined treatment with osimertinib and a PI3K inhibitor reversed the resistance. [51-54].

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
		agents that target mTOR, including dual mTORC1/mTORC2 inhibitors and dual PI3K/mTOR inhibitors, are currently in preclinical and clinical development. [47-49]. The p110-alpha inhibitor alpelisib has been approved for the treatment of men and postmenopausal women with PIK3CA-mutated, hormone receptor positive, Her2 negative advanced or metastatic breast cancer who experience disease progression on or following an endocrine-based therapy. [50].	
TP53 G266V	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [55]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [56-58]. Expression of p53 in normal cells is low; however, TP53 alterations, including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [59-63]. TP53 alterations are believed to be early events in NSCLC, preceding lymph node metastasis. [64]. TP53 mutation and expression of p53 have been correlated with the lung squamous cell carcinoma subtype, and p53 expression in lung squamous cell carcinoma has also been associated with disease stage and higher grade tumors. [65-68]. TP53 mutation has been associated with PD-L1 expression and T-cell infiltration in lung adenocarcinoma samples. [69-73]. TP53 mutations have been significantly associated with the development of distant metastases after diagnosis in early-stage NSCLC in a cohort of 759 patients. [74].	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [75-77]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical cancer models with deficiency of p53 function. [78-80]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [81-86].	Mutations in TP53 may increase resistance to ionizing radiation therapy. [87,88].
EGFR Amplification	The presence of an EGFR abnormality (mutation, amplification, or overexpression) can result in an overabundance or overactivity of Egfr protein, which can lead to excessive proliferation. [1].	EGFR amplification or increased copy number may result in elevated Egfr protein expression and thus predict sensitivity to Egfr targeted therapies. The Egfr tyrosine kinase inhibitors (TKIs) erlotinib, afatinib, gefitinib, osimertinib, and dacomitinib have been approved for the treatment of non-small cell lung carcinoma (NSCLC) with	EGFR amplification has been reported as an acquired alteration in 15.5% (16 /103) of EGFR exon19del or L858R-mutant NSCLC patients who were treated with Egfr TKI and in 9.7% (3 /31) of patients treated with Egfr TKI in combination with bevacizumab. [99].

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
		specific EGFR mutations; however, only modest clinical benefit for gefitinib or erlotinib has been reported in patients harboring EGFR amplification without concurrent sensitizing mutations. [6,8,12,89-93]. Anti-Egfr monoclonal antibodies are also approved in some indications, including cetuximab, which is an approved therapy for head and neck and colorectal cancer, and panitumumab, which is approved in colorectal cancer. [94-96]. However, molecular analyses of tumor samples from a Phase 3 study in head and neck squamous cell carcinoma revealed that neither Egfr expression nor EGFR amplification predicted response to cetuximab. [97,98]. Studies have reported non-squamous NSCLC patients with metastatic disease and tumors harboring an EGFR exon 19 deletion or L858R mutation to be sensitive to osimertinib, erlotinib, afatinib, gefitinib, dacomitinib, and the combination of erlotinib plus ramucirumab. [2,5-7,11,13,18]. Less common activating EGFR mutations have variable sensitivity to EGFR tyrosine kinase inhibitors. [19].	
CCNE1 Amplification	Increased Cyclin E1 levels have been reported to enhance cell proliferation, which is likely due to its key role in the cell cycle. [100-102]. Scientific studies have reported that a number of cancers show CCNE1 amplification, overexpression of Cyclin E1, or increased Cyclin E1-Cdk2 activity. [103-109]. Increased expression of Cyclin E1 has been reported in non-small cell lung carcinoma (NSCLC) cases as compared with normal tissue, and has been correlated with tumor stage and metastasis; Cyclin E1 has also been shown to enhance cell growth of NSCLC cell lines and to drive the formation of lung tumors in murine models. [110-114].	At present there are no drugs that directly target Cyclin E1. However, tumors with CCNE1 amplification or activating mutations may be sensitive to inhibitors of Cdk2, the protein that Cyclin E1 binds and activates. Cyclin-dependent kinase (Cdk) inhibitors that target a variety of Cdks, including Cdk2, are under investigation in early phase clinical trials. [107,115]. The efficacy of this therapeutic approach has been reported in preclinical studies of breast and ovarian cancer cells with CCNE1 amplification or Cyclin E1 overexpression, in which treatment with Cdk2 inhibitors resulted in decreased proliferation and increased apoptosis in the cancer cells and reduced metastatic colonization in xenograft models. [116,117]. The combination of Cdk2 and either Akt or PI3K inhibitors in cancer cells with CCNE1 amplification has also been reported to be effective in preclinical studies. [118,119]. In addition, preliminary results from a Phase 2 trial of the Wee1 inhibitor adavosertib in patients with platinum-resistant ovarian	CCNE1 amplification has been implicated in resistance to Cdk4/6 inhibition in preclinical models of breast and ovarian cancer. [121,122].

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
		cancer indicated that CCNE1 amplification may be positively associated with response. [120].	
PIK3CA Amplification	PIK3CA mutations are not mutually exclusive with EGFR or KRAS or BRAF mutations, and are associated with increased PI3K signaling and increased activation of Akt. [37,38]. PIK3CA mutations have been associated with activation of PI3K/Akt signaling and colony formation in NSCLC cell lines, and the PIK3CA H1047R activating mutation has been shown to drive tumorigenesis in combination with BRAF V600E in a mouse model of NSCLC. [37,39].	Activating PIK3CA alterations may predict sensitivity to PI3K/Akt/mTOR pathway inhibitors, several of which have been approved in specific cancer indications, including the PI3K inhibitors alpelisib and copanlisib, and the mTOR inhibitors everolimus and temsirolimus. [40-42]. While PIK3CA activating alterations have been suggested to predict sensitivity to the mTOR inhibitors everolimus and temsirolimus, results from clinical studies have been mixed, with several reporting no associations between PIK3CA mutational status and response to therapy. [41,43-46]. Other agents that target mTOR, including dual mTORC1/mTORC2 inhibitors and dual PI3K/mTOR inhibitors, are currently in preclinical and clinical development. [47-49]. The p110-alpha inhibitor alpelisib has been approved for the treatment of men and postmenopausal women with PIK3CA-mutated, hormone receptor positive, Her2 negative advanced or metastatic breast cancer who experience disease progression on or following an endocrine-based therapy. [50].	PIK3CA mutations have been reported in EGFR-mutant NSCLC patients following emergence of resistance to osimertinib and have been associated with osimertinib resistance in preclinical NSCLC models. Combined treatment with osimertinib and a PI3K inhibitor reversed the resistance. [51-54] .

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